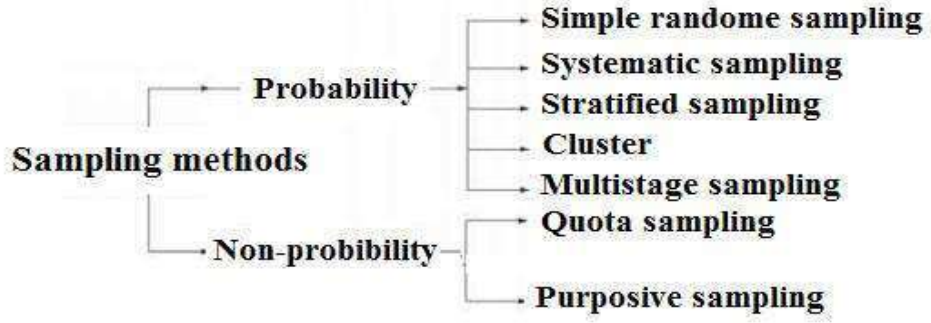
Zoo103 Final Term Handouts 2022

LECTURES 109 to 218

Module# 109 Sampling Methods

- Probability Sampling
- Non-Probability Sampling Probability Sampling
- Random Sampling
- Systematic Sampling
- Stratified sampling
- Cluster Sampling
- Multistage Sampling Non-Probability
- Purposive sampling
- Quota Sampling

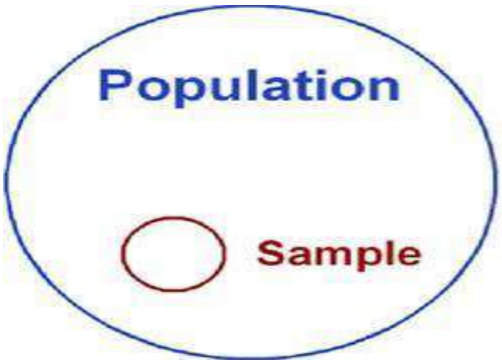


• **Sampling**: It is the process of selecting a representative group from the population. The target population may be as board as all humans but it may be a smaller group as people who are diabetics or only the teenagers.

• **Sample**: A sample is a group of units selected from a population by a defined or simply a part of a statistical population.

• The sample should be representative of the general population. 510% portion of the sample size of total

• **Population**: Population is the total group of individuals from which the sample might be drawn.



• All members of population possess the same basic and defined characteristics, but varying in amount or quality from one member to another.

Module# 110 Random Sampling

• If every individual of target population has an equal chance of selection then it will be called random sampling. Simple random sampling is the easiest form of probability sampling.

• No biasness and expert opinion is required in this method.

Random Sampling

• All the researcher needs to do is assure that all the members of the population are included in the list and then randomly select the desired number of subjects.

Examples:

• Selection of plants from rice fields

• Selection of students from class

• When carrying out a survey, it would be impractical to study a whole population.

Module# 111 Systematic Sampling

• In this sampling method first sample from the population is taken randomly but after that samples are selected systematically at a regular interval.

Examples:

• Shopping Centre

• Selection of plants

• Selection of soil sample

Module# 112 Principles of Systematics

Stratified Sampling

Module# 113 Cluster Sampling

• It is a sampling technique where the entire population is divided into groups, or clusters and a random sample of these clusters are selected.

• Members of one cluster have similarity with each other and also with other clusters; minor differences can be there between different clusters.

• Individual selection from each cluster is random.

Cluster Sampling

• Cluster of plants for study

• Cluster of animals

Cluster Sampling

• It is a sampling technique where the entire population is divided into groups, or clusters and a random sample of these clusters are selected.

• Members of one cluster have similarity with each other and also with other clusters; minor differences can be there between

Module# 114 Multistage Sampling

• It is complicated form of cluster sampling. In this type of sampling we divide our population into groups (cluster) and each group is further divided into sub- groups in such a way that each lower group is adjusted into higher group.

Multistage Sampling

• For example in a survey of tuberculosis prevalence in world, if we divide whole world into continents, then select 10countries from each continent randomly, then 10districts from each country, 10 cities from each district, 10 villages from each city and finally 10 families from each villages randomly. This type of END sampling is multistage sampling.

Module# 115 Principles of Systematics

Non-Probability Sampling

• **Quota sampling:** In this type of sampling we divide the population intro different groups or strata based on some specific criterion but selection of individual from each group is not random.

• **Purposive sampling:** In this technique researcher selects sample based on their knowledge and professional judgment. Purposive sampling can also be known as judgmental, selective or subjective sampling.

For example purchase of car from market based on expert opinion or selection of staff for biology teaching in a college by a panel of subject experts.

Module# 116 Construction of Phonogram

• A tree diagram depicting taxonomic relationships among organisms based on overall similarity of many characteristics without regard to evolutionary history

Rules

•Start from lowest value

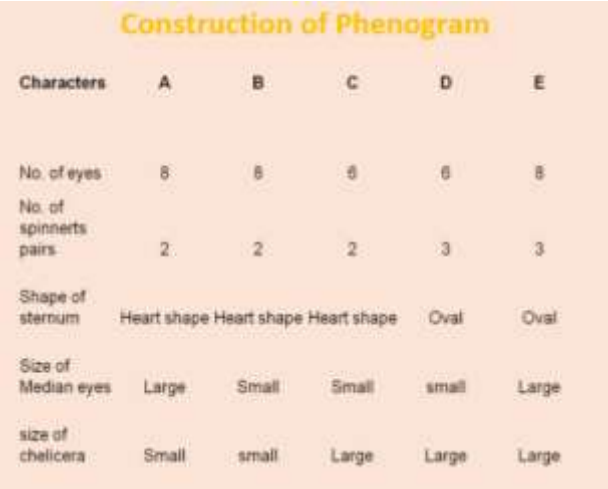
•Try to make maximum separate clusters

•Do not repeat

•Incase of same number multiple time, you can chose any combination

• Value of previous similarity should be higher than afterwards

Construction of Phonogram

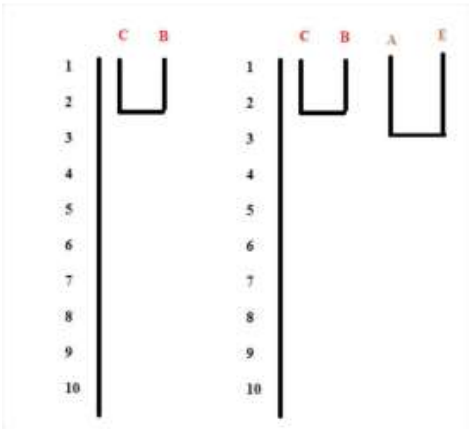


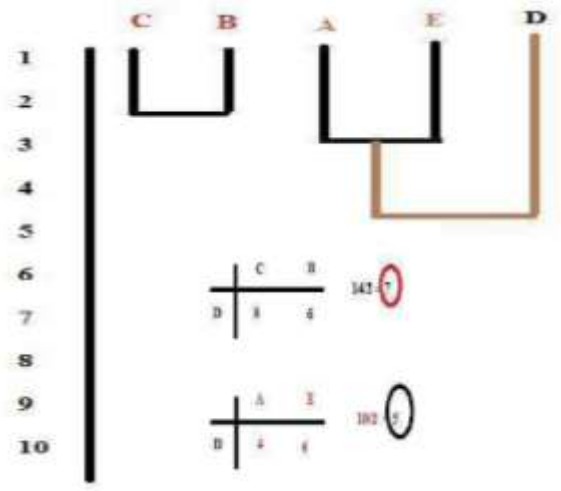
	A	B	C	D	E
A	x				
B	9	x			
C	5	2	x		
D	4	6	8	x	
E	3	5	7	6	x

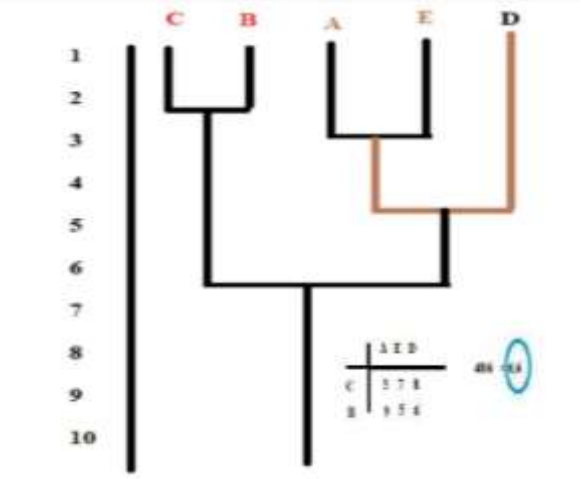
	A	B	C	D	E
A	x				
B	9	x			
C	5	2	x		
D	4	6	8	x	
E	3	5	7	6	x

Module# 117 Construction of Phenogram

	A	B	C	D	E
A	x				
B	9	x			
C	5	2	x		
D	4	6	8	x	
E	3	5	7	6	x

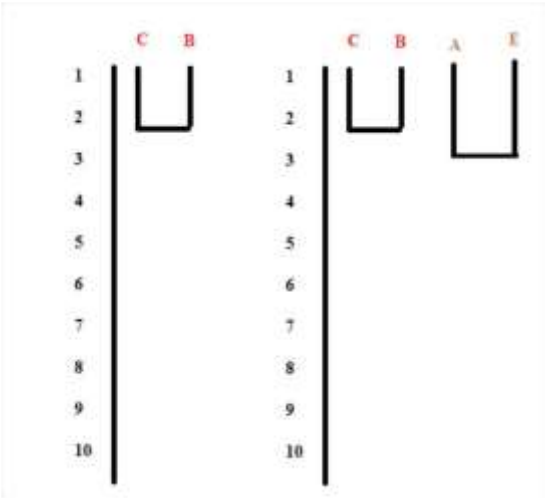


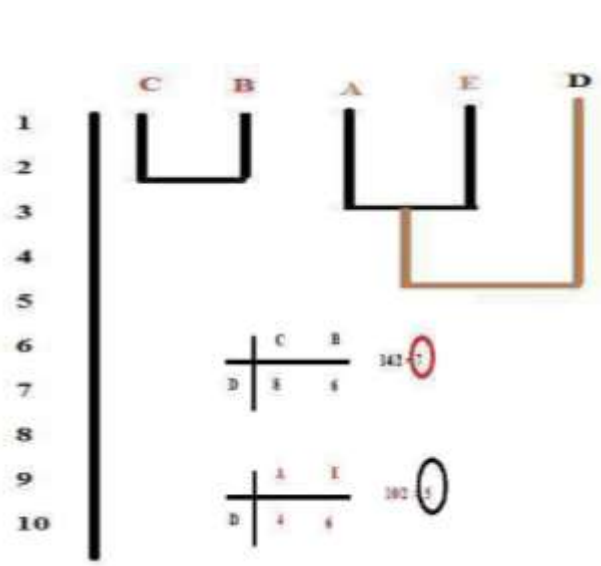


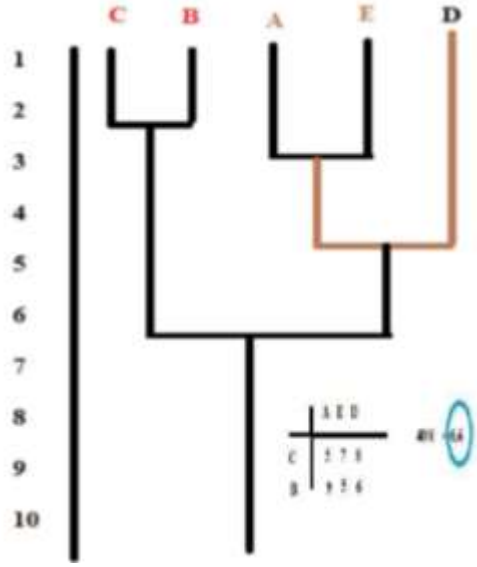


Module# 118 Construction of Phenogram

	A	B	C	D	E
A	x				
B	9	x			
C	5	2	x		
D	4	6	8	x	
E	3	5	7	6	x







Module# 119 Describing a New Species

•Describing new species is not something that is done by non - taxonomists.

• To describe a new species, you need to have the appropriate credentials, experience, and knowledge to do so.

• Assuming you have all describing a New Species

• Carrying out the necessary measurements (scales, morphological size, characteristics, etc.) needed in a description.

• Record location, habitat, and other info such as DNA Describing a New Species

• Evidence of comparative studies demonstrating that your species is indeed new or undescribed (this can take a long time to do properly).

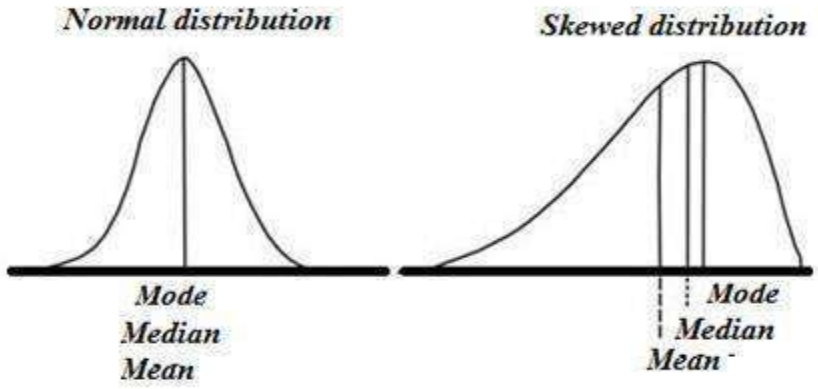
• Finding the right journal for publishing your description.

• Zootaxa, ZooKeys

Module# 120 Normal and Skewed Data

• In normal data values of means, median mode are equal but this not the case for asymmetric or skewed data.

• Normal data follow a bell shape curve. In skewed distribution one tail is larger than other as in above figure.



• In normal data most of the individual values are close to the mean of the data. However, in skewed t the hat difference between individual values and mean values is higher.

• Extreme values are almost negligible in normal data but higher

• Values of standard deviation are less for normal data compared to skewed data.

• Measure of central tendency for normal data is mean but it is median for skewed data.

Module# 121 Parametric Vs Non - Parametric Tests

• Parametric tests are used for Normal data

• T – test • Z – test • F – test • ANOVA • Regression • Correlation

• Non - parametric tests for skewed data. • Wilcoxon Signed Rank Test • Mann Whitney U Test

• Kruskal - Wallis test • Principal Component Test

Parametric Test	Non-Parametric test
Make assumptions about the population	Does not make assumptions
Uses normal probabilities	Distribution is arbitrary
Uses mean value for central tendency	Uses median value for central tendency
Require previous knowledge about the population	Does not require previous knowledge about the population

Friedman Tests Parametric Vs Non - Parametric Tests • for measuring the degree of association between two quantitative variables, **Pearson's coefficient of correlation** is used in the parametric test, while **Spearman's rank correlation** is used in the nonparametric test.

Module# 122 Types of data Distribution

• Normal Distribution

• Binomial Distribution

• Poison Distribution

Normal Distribution

• Data which is normally distributed and follow the normal curve is called normal data. It can be continuous and discrete in value.

Binomial Distribution

• Data in which there are only two outcomes from an event and numbers of events are predefined is called binomial distribution.

For example in five test match series, outcomes of each match are two i.e. to win or lose. Outcome of the event is a discrete number.

Poison Distribution

• The data distribution in which total numbers are not predetermined and some time there number can reach to infinite. These events occur in nature rarely. Outcome of such events is complete discrete value not in decimal points.

For example number of deaths in Lahore per hour. Number of deaths can be infinite if some unusual event occurs.

Module# 123 Measure of Central Tendency and Measure of Variability

Measure of central tendency

• It provide information about average value of a data set.

• Mean, Median and Mode are commonly used measures of central tendency.

• Data set has only one and only one Mean.

• Value of mean is influenced by all observations in the data set.

Measure of variability

• Mean deviation

• Variance

• Standard deviation

• Standard error of mean

• Range

• Less vs high value

Module# 124 Z - test

• Z - test is used to compare the means of two samples when sample size is greater than 30 (n >30).

$$Z = \frac{\overline{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Z = X (BAR) – U(macro) / SE

• This test like t - test assume a normal distribution of the data.

• This test is most useful when the standard deviation is known.

Z - Test

• Test the hypothesis that there is no difference in the calculated sample mean with given population means.

$$z = \frac{\overline{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

• Population mean: μ = 100, n= 55, Sample mean: and = 12

• By putting the values in formula:

• Z - calculated= - 2.7

Z - Test

• Critical value at 95% confidence interval for Z - test is ±1.96. Z Calculated is not within the acceptable range so our null hypothesis is rejected.

Module# 125 ANOVA

• Analysis of variance (ANOVA) is a technique used to compare means of more than two samples.

• This technique can be used only for numerical data as it is a parametric test."

Assumptions

• Number of samples must be more than two.

• Data should be normally distributed.

• Samples must be independent.

• Homogeneity of variances.

Examples

• A group of psychiatric patients are trying three different therapies: counseling, medication and biofeedback. You want to see if one therapy is better than the others.

• Students from different colleges take the same exam. You want to see if one college outperforms the other.

Module# 126 ANOVA

Source of variation	SS	DF	MS	F-value
Between group variation				
Within group variation				
Total				

• SS (total) [Sum of square] = (Each observation – Grand mean) 2(whole Square)

• SS (within) = ((Each observation – Group mean) 2 (whole Square)

• (between) SS = (Group mean Grand mean) 2 (whole Square) × Number of observation in each group

• DF (between) = Total number of Groups – 1

• DF (within group) = Total number of Observations – Total number of Groups

Module# 127 One - Way ANOVA

• A one - way ANOVA uses one independent variable, while a Two - way ANOVA uses two independent variables.

• As a crop researcher, you want to test the effect of three different fertilizer mixtures on crop yield. You can use a one - way ANOVA to find out if there is a difference in crop yields between the three groups.

• You are interested to compare the effectiveness of three medicines in treating cancer patients

• Comparison of four feeds on weight of fish

Replicates	Lahore	Sialkot	Sargodha
1	8	10	8
2	14	14	6
3	3	11	5
4	7	9	1
Mean	**8**	**11**	**5**

Grand Mean = 8+11+5/3 = 8

• SS (total) [Sum of square] = (Each observation – Grand mean) 2 (whole Square)

• SS (Total) = (8 - 8) 2 (whole Square) + (14 - 8) 2 (whole Square) + (38) 2 (whole Square) + (7 - 8) 2 (whole Square) + (10 - 8) 2 (whole Square) + (14 - 8) 2 (whole Square) + (118) 2 (whole Square) + (9 - 8) 2 (whole Square) + (8 - 8) 2 + (6 - 8) 2 (whole Square) + (6 - 8) 2 + (5 - 8) 2 (whole Square) + (1 - 8) 2 (whole Square)

• **SS (Total) = 174**

SS (within) = ((Each observation – Group mean) 2

• SS(within) = (8 - 8) 2 +(14 - 8) 2 + (38) 2 +(7 - 8) 2 +(10 - 11) 2 +(14 - 11) 2 +(11 - 11) 2 +(9 - 11) 2 +(8 - 5) 2 +(6 - 5) 2 +(5 - 5) 2 +(1 - 5) 2

•SS (with in) = 102

Module# 128 One - Way ANOVA

• **SS (between)** = (Group mean Grand mean) 2 × Number of observation in each group

• SS (between) = [(8 - 8) 2 × 4] + [(118) 2 × 4] + [(5 - 8) 2 × 4] = 72

• **DF (between)** = Total number of groups – 1

• DF (between) = 3 - 2 1 =

Source of variation	SS	DF	MS	F-value
Between group variation	72	2	36	3.17
Within group variation	102	9	11.33	
Total	174	9+2 = 11	-	

•DF (within group) = Total number of observation – Total number of groups

• DF (within group) = 12 - 3 = 9

MS (between)

• It can be calculated by dividing the

SS (between group) with DF

(between group)

• 72 /2 = 36

MS (within)

It can be calculated by dividing the SS (within group) with DF (within group)

• 102/9 =11.33

F - Calculated

•3.11

F -Tabulated

• Comparison of both • Decision about hypothesis

Module# 129 Two - Way ANOVA

• In two - way ANOVA there are two independent variables and one dependent variable.

• In one - way ANOVA there was only one independent variable

Two - Way ANOVA Assumptions

• The population must be close to a normal distribution.

• Samples must be independent.

• Population variances must be equal.

• Groups must have equal sample sizes.

• One could use a two - way ANOVA to understand whether there is an interaction between gender and educational level on test anxiety amongst university students, where gender (males/females) and education level (undergraduate/postgraduate) are your independent variables, and test anoxia ty is your dependent

• For example, a two - way ANOVA allows a company to compare worker productivity based on two independent variables, such as department and gender.

• It is utilized to observe the interaction between the two factors. It tests the effect of two factors at the same time.

• The effective life (in hours) of batteries is compared by **material** type (1, 2 or 3) and operating **temperature**: Low (- 10 ˚C), Medium (20˚C) or High (45˚C). Two - Way ANOVA

• Twelve batteries are randomly selected from each material type and are then randomly allocated to each temperature level.

Module# 130 T - test

• T - test is appropriate whenever you want to compare the means of two samples.

• This test is used to find out whether the means of two samples are statistically different from each other or not.

• Data should be normally distributed.

• Samples sho uld be taken randomly from the population

• Sample size should be 30 or less than 30.

• All randomly collected samples should have the same variances (within the limits of sampling error).

• Standard deviation may or may not be known.

• Independent t - test/two sample t-test

• Paired t – test

Module# 131 Independent t - test

•For two sample t - test, the two populations must be independent

• It is not necessary that sample size should be same for both samples.

• Example s: Test the hypothesis that mea ns heights (cm) of plants randomly selected from two wheat fields are same. Independent t - test

• For two sample t - test, the two populations must be independent

• It is not necessary that sample size should be same for both samples.

• Example s: Test the hypothesis that means heights (cm) of plants randomly selected from two wheat fields are same.

Data of field I (X_1)= 10, 15, 20. 25 and 30.

Data of field II (X_2)= 4, 8, 10. 15 and 10.

Null hypothesis: $\bar{X}_1 = \bar{X}_2$

Alternative hypothesis is $\bar{X}_1 \neq \bar{X}_2$.

Formula:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

X_1	X_2	X_1^2	X_2^2
10	4	100	16
15	8	225	64
20	10	400	100
25	15	625	225
30	10	900	100
$\sum X_1 = 100$	$\sum X_2 = 47$	$\sum X_1^2 = 2250$	$\sum X_2^2 = 505$

$$\sigma_1^2 = \frac{\Sigma(X_1 - \bar{X}_1)^2}{n-1}$$

$$= \frac{\Sigma X_1^2 - \frac{(\Sigma X_1)^2}{n}}{4} = \frac{2250-2000}{4} = 62.5$$

$$\sigma_2^2 = \frac{\Sigma(X_2 - \bar{X}_2)^2}{n-1}$$

$$= \frac{505 - \frac{(47)^2}{5}}{4} = \frac{505-441.8}{4} = \frac{63.2}{4} = 15.8$$

$$t = \frac{20-9.4}{\sqrt{\frac{62.5}{5} + \frac{15.8}{5}}}$$

$$= \frac{10.6}{\sqrt{12.5+3.6}} = \frac{10.6}{\sqrt{15.66}} = \frac{10.6}{3.95} = 2.68$$

Null hypothesis: $\bar{X}_1 = \bar{X}_2$

t = 2.68 (calculated value)

As both samples are independent so degree of freedom will be n-1 (from sample X_1) and n-1 (from sample X_2).

5-1 = 4 (X_1) and 5-1 = 4 (X_2)

Degree of freedom: 4+4 =8

Tabulated value/standard value/critical value = 2.3

Result: As our calculated value is lower than tabulated value so our null hypothesis is accepted.

Module# 132 Paired t - test

X	Y	d	d^2
9	12	-3	9
12	14	-2	4
13	19	-6	36
20	25	-5	25
5	10	-5	25
		$\sum d = -21$	$\sum d^2 = 99$

$$\mathbf{t} = \frac{\bar{\mathbf{d}}}{\sqrt{\frac{\sigma^2 \mathbf{d}}{\mathbf{n}}}}$$

$$\sigma^2 \mathrm{d} = \frac{\sum(\mathrm{d} - \bar{\mathrm{d}})^2}{\mathrm{n} - 1}$$

$$= \frac{\sum \mathrm{d}^2 - (\bar{\mathrm{d}})^2}{\mathrm{n}-1}$$

$$= \frac{99 - 17.64}{4} = \frac{81.36}{4} = 20.34$$

$$\bar{\mathrm{d}} = \frac{-21}{5} = -4.2$$

$$t = \frac{-4.2}{\sqrt{\frac{20.34}{5}}} = \frac{-4.2}{\sqrt{4.068}} = \frac{-4.2}{2.016} = -2.08$$

T calculated: -2.08

T tabulated $=2.77$

As observation were paired so degree of freedom (n-1)= 4.

Our null hypothesis is rejected as T- calculated is smaller than T-tabulated.

Module# 133 Standard Deviation

- It is positive square root of variance.
- Standard deviation "provides a sort of average of the differences of all scores from the mean.
- It is a measure of the dispersion of scores around the mean.

$$SD = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

• Generally, a low standard deviation mea ns that a set of scores is not very widely dispersed around the mean.

• While a high standard deviation indicates that the scores are more widely dispersed

• Its advantage over variance is that its physical unit are same as that of individual observations

• The standard error is also a measure of spread.

• It is represented by SE.

$$SE = \frac{\sigma}{\sqrt{n}} \quad \text{or} \quad SE = \frac{SD}{\sqrt{n}}$$

• Higher value of SE indicates higher spread of the data.

• The difference in the standard error and standard deviation is that standard error uses sample data but standard deviations use population data.

• SE tells us that how far our sample mean deviate from Standard Deviation population means.

Module# 134 Standard Deviation

• The standard deviation is a measure of the spread of scores within a set of data.

• Usually, we are interested in the standard deviation of a population.

The **sample standard deviation formula** is:

$$s = \sqrt{\frac{\sum(X - \bar{X})^2}{n-1}}$$

where,

s = sample standard deviation
$\sum$ = sum of...
$\bar{X}$ = sample mean
n = number of scores in sample.

• However, as we are often presented with data from a sample only, we can estimate the population standard deviation from a sample standard deviation.

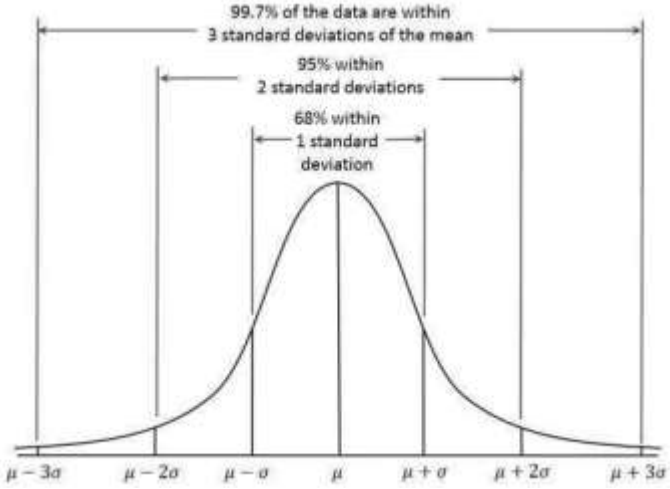
• The standard deviation is a measure of the spread of scores within a set of data.

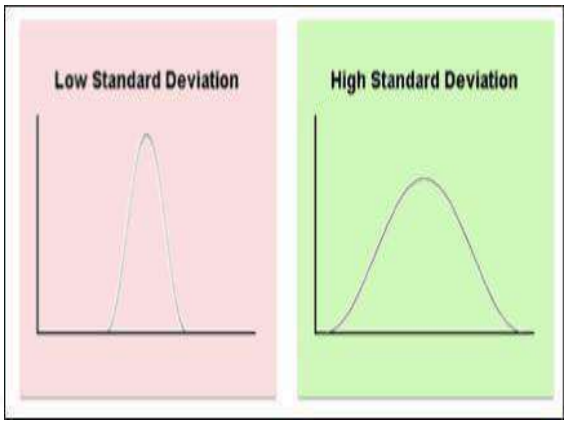
• A **sample standard deviation** is a statistic. This means that it is calculated from only some of the individuals in a population.

• The **population standard deviation** is a parameter, which is a fixed value calculated from every individual in the population.

Relationship between Variance and SD

• The standard deviation is the square root of the variance. The standard deviation is expressed in the same units as the mean is, whereas the variance is expressed in squared units,





Module# 135 Introduction to Spiders

- Most abundant natural predators (Maloney et al., 2003)

• Ranked 7th in animal diversity (WSC, 2018)

• More than 46000 species

• Strictly carnivorous

• Cause wasteful killing

• Show functional and numerical response

• Best survivors under food shortage

• Prey switching

• Cannibalistic

• Top down effect

• Indirect killing

Module# 136 Introduction to Spiders

• Ground runners • Stalkers • Web builders

• Bio - control agents

• Model organism for research

• Spiders silk

• Spider venom peptides

Module# 137 Morphology of spider Leg

• Spiders typically have eight walking legs (insects have six)

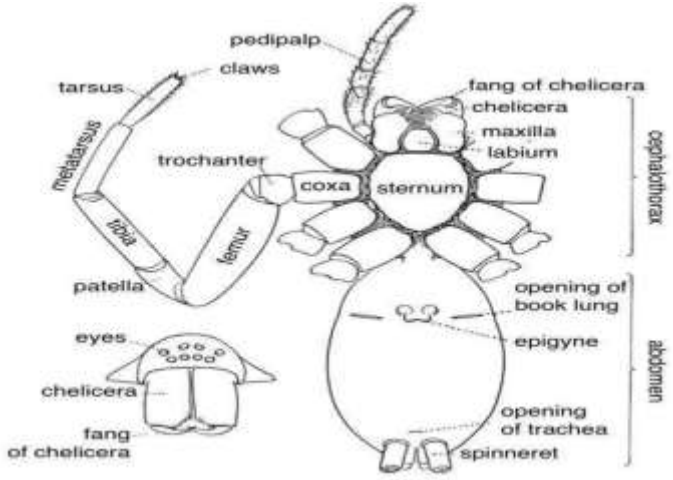
• They do not have antennae

• The pair of app endages in front of the legs are the Pedi palps (or just palps)

• Spiders' legs are made up of Morphology of spider Leg seven segments

• The tip of the tarsus bears claws, which vary in number and size

• Spiders that spin webs typically have three claws, the middle one being small





• Hunting spiders typically have Morphology of spider Leg only two claws

• Starting from the body end, these are the: • Coxa • Trochanter • Femur • Patella • Tibia

• Metatarsus • Tarsus.

• Since they do not have antennae, spiders use specialized and **sensitive setae** on their legs to pick up scent, sounds, vibrations and air currents.

• Some spiders, such as not have claws. The Australia crab spiders, do not have claws.

• The pedipalps have only six segments: the metatarsus is missing.

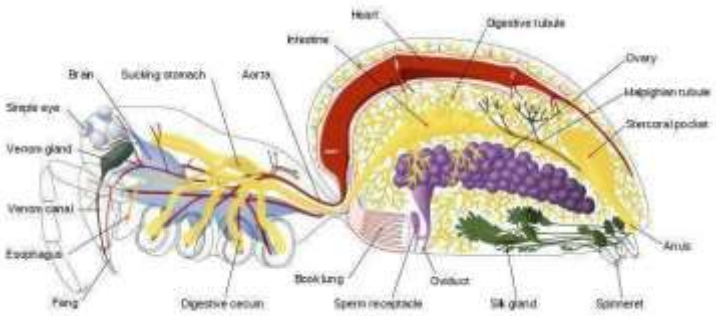
• In adult males, the tarsus of each palp is modified for mating (variously called a palpal pulb, palpal organ or copulatory bulb)

Module# 138 Morphology of spider's Abdomen

• The abdomen is also known as the **opisthosoma**.

• On the ventral side of the abdomen are two hardened plates covering the book lungs.

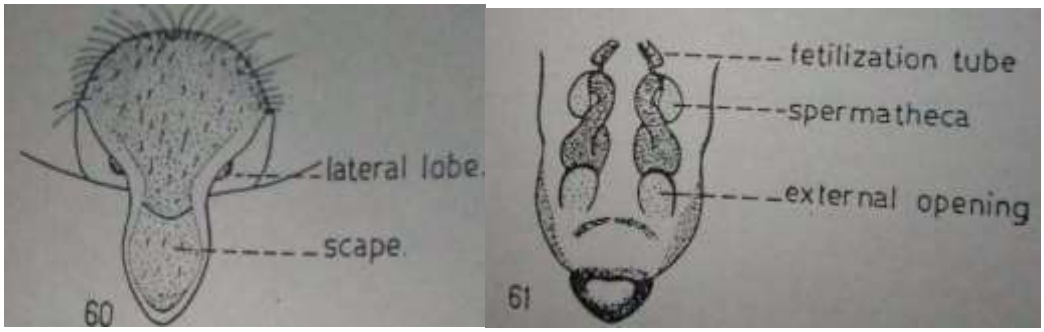
• Females genital organ is also present on ventral side of the abdomen.



• It is called **epigyne**.

• These are called the **epigastric plates**

• In many cases epigyne is provided with a tongue like sclerotic projection called **scape**.



• Scape often have paired lateral lobes as in Neoscon Tikadar, 1982. The Funa of India

• Epigine has two external openings which lead to sperm thecae inside and paired of narrow **curved tubes called** **fertilization tube**

• Scape often have paired lateral Tikadar, 1982. The Funa of India lobes as in Neoscon

• A fold, known as the epigastric furrow , separates the region of the book lungs and epigyne .

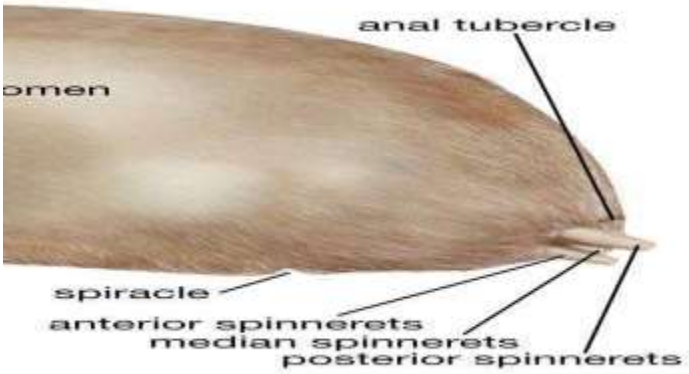
• In the middle of this furrow is the opening of the oviduct (in females) and at either end are the lung slits.

• On posterior side of the abdomen there is conical anal tubercle that bear three pairs of spinnerets ventrally.

• On ventral side there are two pairs of book lungs.

•

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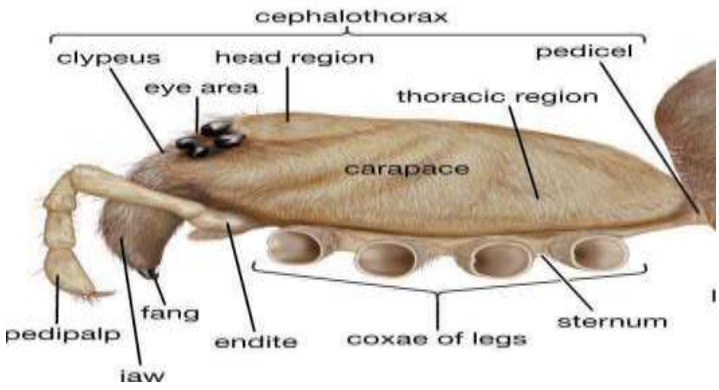
Module# 139 Morphology of spider's Cephalothorax

• Cephalic region has 6 - 8 eyes

• Eyes or black or diurnal

• White or nocturnal

• If only one type of eyes (homogenous)



• If both type of eyes or present (heterogeneous) usually arranged in two rows

• Each row usually contain 4 eyes. • Rows of eyes are frequently curved.

• Anterior median/Anterior lateral • Posterior median/Posterior lateral

• Cephalic area of eyes is called **ocular area.**

• Area of four median eyes is called **ocular quad.**

• The area between anterior row of eyes and anterior margin of carapace is **clypeus.**

• The depression in the middle of thorax is called **thoracic groove or fovea**.

• Appendages • Chelicerae/Fangs

• Padipalp (2nd pair of appendages) • Papl has six parts/metatarsus missing.

• In males tarsus/modified copulatory organ • Ventral side is sternum

• Oval/heart shape

Module# 140 Morphology of spider's Spinnerets

• The abdomen of spiders has modified pairs of movable telescoping organs called **spinnerets**, which produce silk.

• The suborder **Mesothelium** is unique in having only two types of s ilk glands – thought to be the ancestral condition.

• Depending on the complexity of the species, spiders will have two to eight spinnerets, usually in pairs.

• Behind each spinneret visible on the surface of the spider lies a gland.

• Each spigot has a valve that controls the thickness and speed of the extruded material. As the spigots pull the **fibroin protein molecules** out of the ducts and extrude them into the air, the molecules are stretched out and linked together to form long strands.

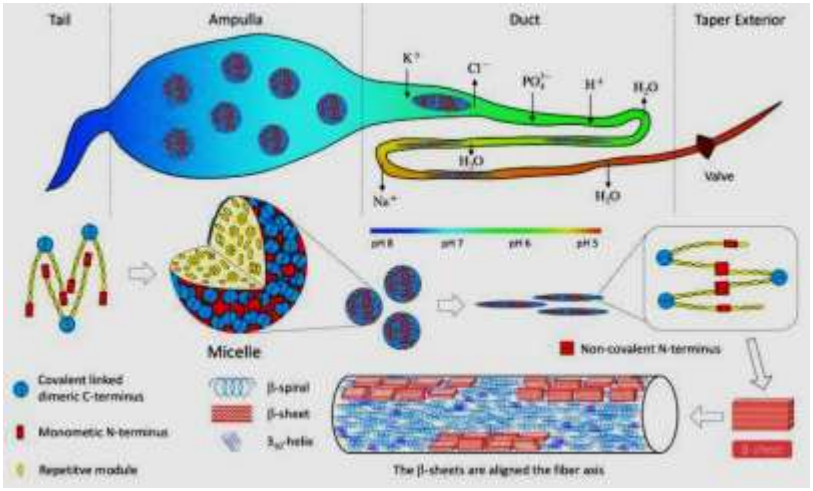
• The **spinnerets** wind these strands together to form the sturdy silk fiber.

• Most spiders have multiple silk glands, which secrete different types of silk material optimized for different purposes.

• Spider silk is incredibly strong and flexible. Five times as strong as an equal mass of **steel** and twice as strong as an equal mass of **Kevlar**.

• Elasticity

• Size Temperature/PH/uses



Module# 141 Introduction to Scorpions

• Scorpions are predatory arachnids of the order Scorpions.



• They have eight legs, and are easily recognized by a pair of grasping pincers and a narrow, segmented tail, often carried in a characteristic forward curve over the back and ending with

• Scorpions primarily prey on insects and other invertebrates, but some species take vertebrates.

• They use their pincers to restrain and kill prey.

• The vast majority of species do not represent a serious threat to humans, and healthy adults usually do not need medical treatment after being stung.

• Poisonous scorpions • Habitat/desert/Mountains • Collection/UV/Turning stone/pitfalls

Module# 142 Introduction to Scorpions

• Scorpion bites are highly painful • Venom peptides

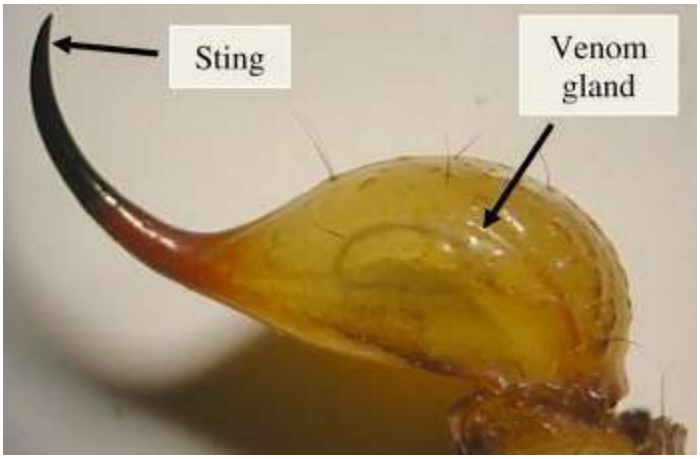
• Type/Long chain/Short chain • Medical importance/Buthidae

• Treatment of diseases • Cancer treatment

• Venom price • Venom collection

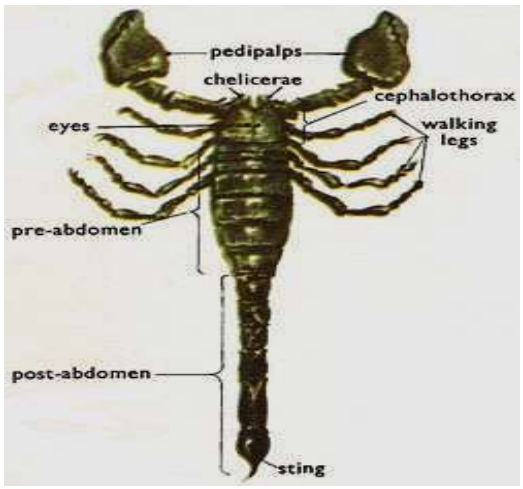
• How to collect venom • Effect of temperature on venom production

• Amount of venom • Method of venom collection



Module# 143 Morphology of Scorpions

• Cephalothorax • Padipal • Chelicera • Abdomen • Walking legs • Sting



• Body is covered with a cuticle consisting of chitin and tanned proteins • 8.5-23mm

• The opisthosoma is subdivided into a broad anterior portion, the mesosoma or pre-abdomen, and a narrow tail-like posterior, the metasoma or post-abdomen.

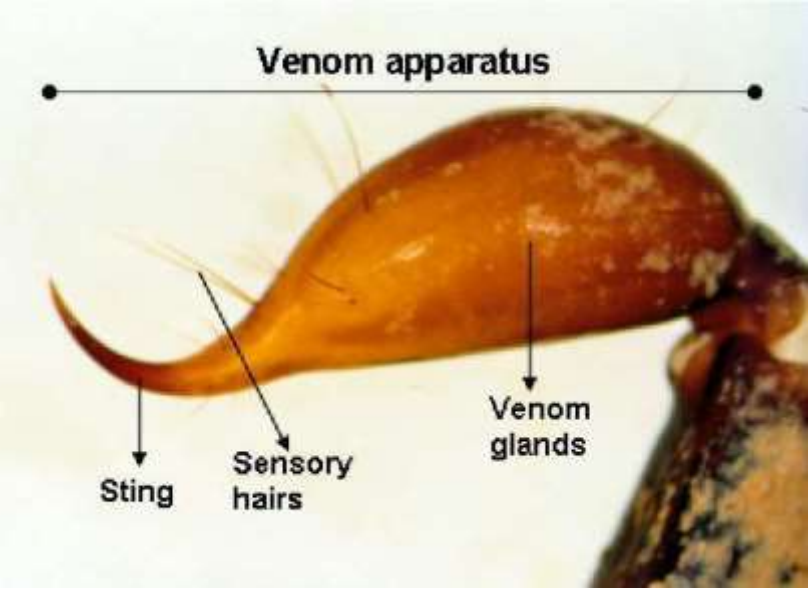
• Most species are nocturnal. • Feeding habit/cannibalism

• The telson includes the vesicle, which contains a symmetrical pair of venom glands.

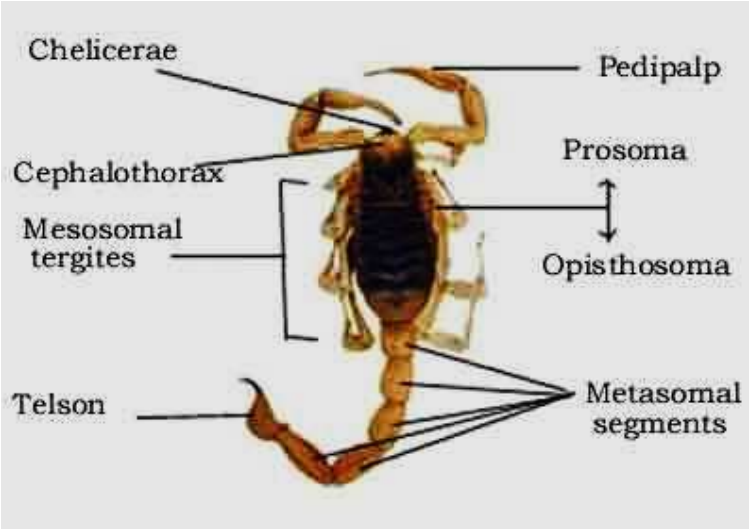
• Externally it bears the curved stinger, the hypodermic aculeus, equipped with sensory hairs.

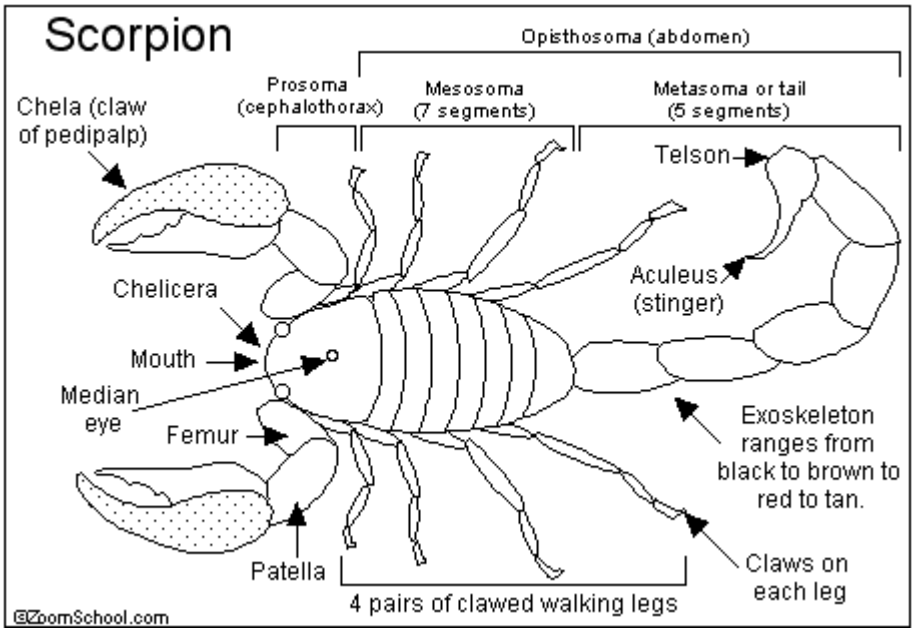
• Abdominal segments these segments have keels, setae and bristles which may be used for

• Each of the venom glands has its own duct to convey its secretion
along the aculeus from the bulb of the gland to immediately sub terminal of the point of the aculeus.



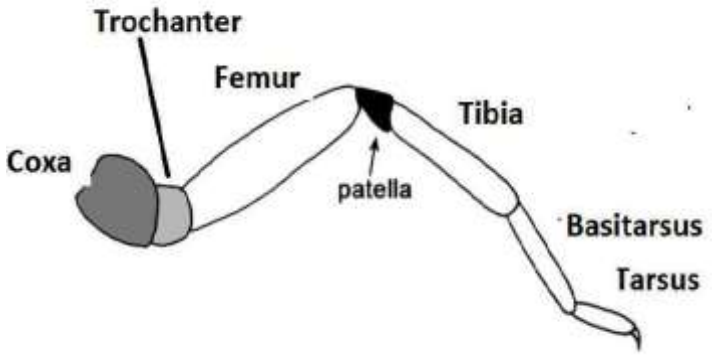
• Parts of scorpion body





Module# 144 Morphology of Scorpion Leg

• Four pairs of legs • each leg is terminating in a pair of claws.



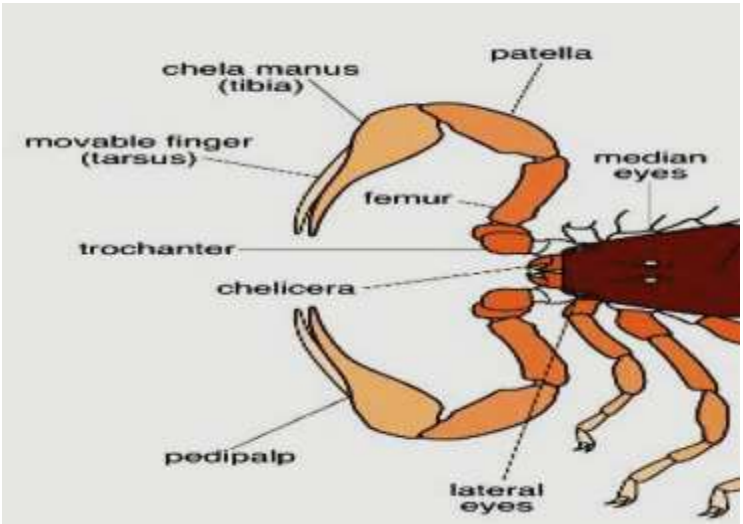
• A series of trichobothria, the number and positions of which are taxonomically informative, is present on the pedipalp femur, patella, and tibia.

• Each chelicera comprises three segments, including a chela manus with fixed finger and a movable finger.

• Each pedipalp comprises a coxa, trochanter, femur, patella, chela manus with fixed finger (tibia), and movable finger (tarsus).

• Setae are thick hairs which cover the legs, body, and pedipalps (leg-like mouthparts).

• They are sensitive to the touch.



• Also located on the pedipalps are smaller hairs known as trichobothria, which sense air vibrations.

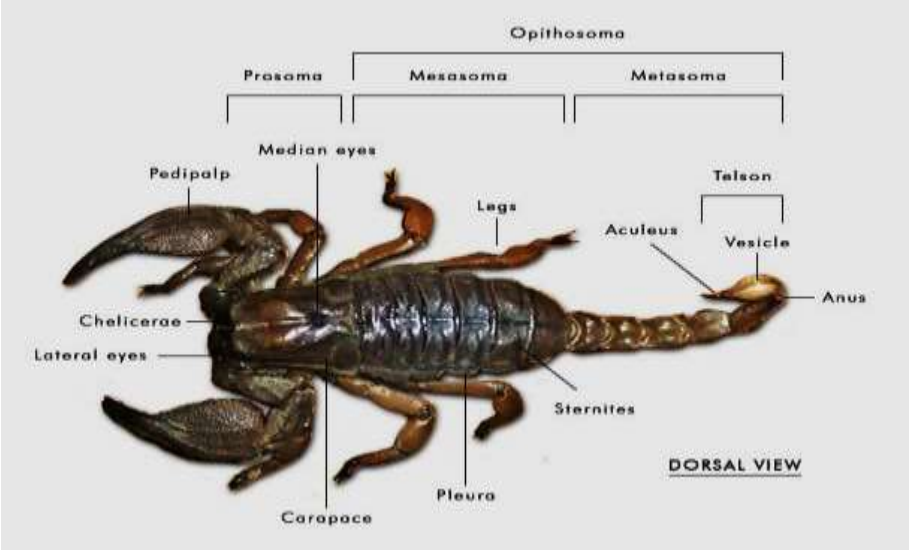
• Ground vibrations are sensed by other organs located on the tip of the legs.

Module# 145 Morphology of Scorpion Telson

• The telson is present in Metasoma.

• Metasoma is also called as post- abdomen.

• The post abdomen includes five metasomal segments, the last of which contains the anus and bears the telson.



• The telson consists of the vesicle, which holds a pair of venom glands, and the hypodermic aculeus, the venom-injecting sting.

• Each of the venom glands has its own duct to convey its secretion
along the aculeus from the bulb of the gland to immediately sub terminal of the point of the aculeus, where each of the paired ducts has its own venom pore

• **An extrinsic muscle system** in the tail moves it forward and propels
and penetrates with the aculeus, while an **intrinsic muscle system**
attached to the glands pumps venom

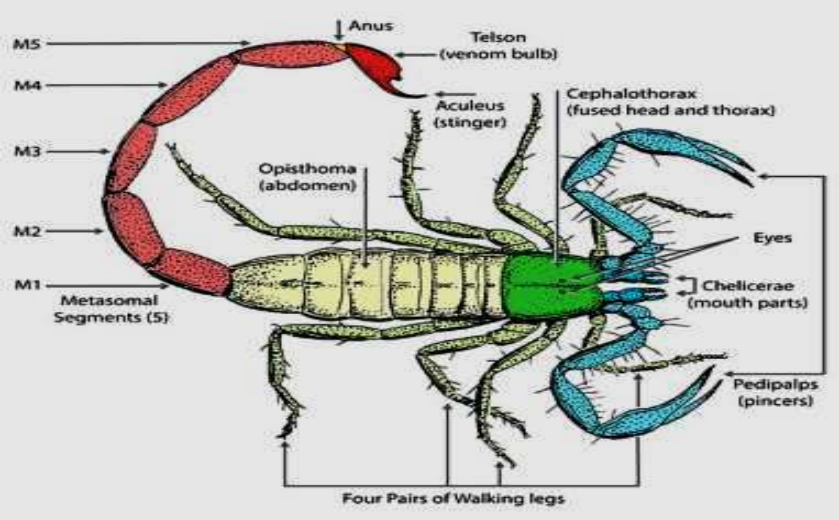
• All scorpion species possess venom, which is used to kill or paralyze prey and as a defense against predators.

Module# 146 Morphology of Chelicerae and Abdomen

• The chelicerae of scorpion is present in cephalothorax region.

• They are scorpion's jaws.

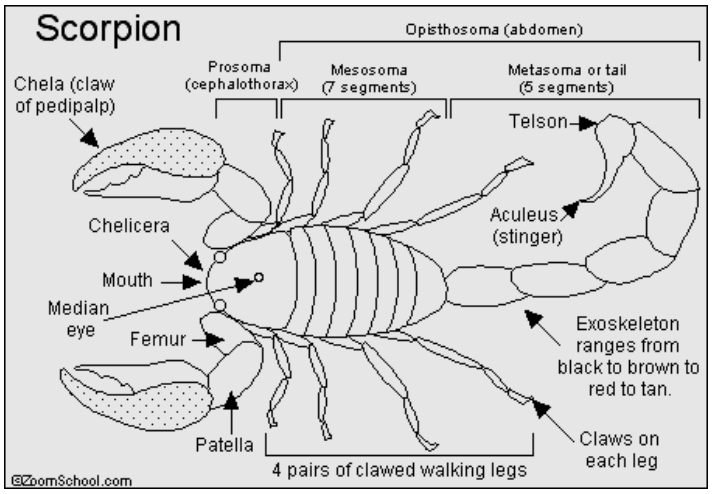
• Each chelicera comprises three parts a coxa, tibia which has a fixed finger, and the tarsus forming a moveable finger.



• Within the abdomen are the scorpion's respiratory system, which is a set of thin plates called **book lungs**.

• A spiracle leads to each pair of book lungs on each segment.

• The opisthosoma has eight legs arranged into **four pairs**.



Module# 147 Accumulation Curve

• The species discovery curve or species accumulation curve is a graph recording the cumulative number of species of living things recorded in a particular environment as a function of the cumulative effort expended searching for them.

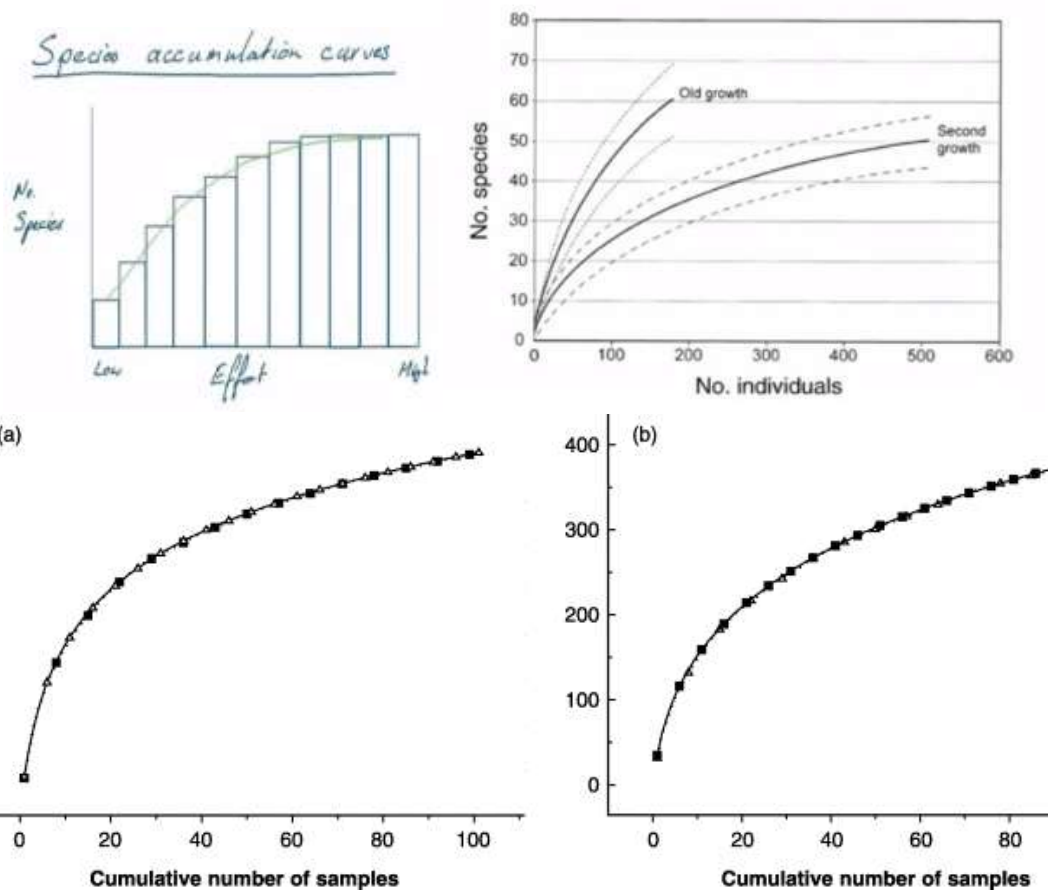
• **Types of accumulation curve**

• Importance of accumulation curve

• Accumulation curve vs trapping effort

• Accumulation curve vs trapping method

• Accumulation curve vs biasness



Module# 148 Calculation of Allelic and Genotypic Frequency

Hardy-Weinberg Equilibrium Model

- P = frequency of Allele A
- q = Frequency of Allele a
- P2 = Frequency of AA genotype
- 2pq= Frequency of Aa genotype
- q2 = Frequency of aa genotype

$$p + q = 1$$

$$(p + q)^2 = 1$$

$$p^2 + 2pq + q^2 = 1$$

	AA	Aa	aa	
Individuals	70	80	50	(200)
Genotypic Array	70/200 = 0.35	80/200 = .40	50/200 = 0.25	= 1
"A' alleles	140	80	0	220/400 = 0.55
'a' alleles	0	80	100	180/400 = 0.45

Module# 149 Calculation of Allelic and Genotypic Frequency

Hardy-Weinberg Equilibrium Model

• In a population there are 320AA individuals, 160 heterozygous (Aa) and 20aa individuals.

• Calculate the genotypic and allelic frequencies.

ALLELIC FREQUENCIES

A 320+320+160/1000 =0.8

a 160+40/1000 =0.2

$$\text{Alleles} = A, a$$

$$\text{Genotypes} = AA, Aa, aa$$

$$\text{Frequency of allele A:} \quad f(A) = f(AA) + 1/2\, f(Aa)$$

$$\text{Frequency of allele a:} \quad f(a) = f(aa) + 1/2\, f(Aa)$$

Genotypes	AA	Aa	aa
Number of individuals	320	160	20
Genotypic frequencies	320/500=0.64	16/500=0.32	20/500=0.04

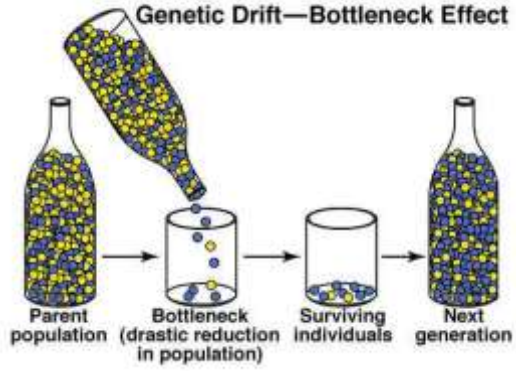
Module# 150 Genetic Drift and its impact on Allele Frequency

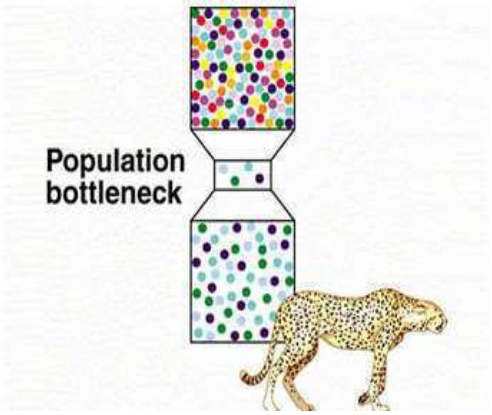
• Change in allege frequency which occur by chance. • Loss in alleles

• Loss in genetic diversity • Lost genes for ever • Bottle neck effect

• Unique genes

• Examples





Module# 151 Principles of Systematics

Concept of fitness

- Survival+ reproductive success
- Relative fitness
- Ideal population
- Maximum value of fitness
- Genetic load/Genetic burden
- Population with high load
- Population with low load
- Variability in fitness

Genetic Load = Max. fit individual - Ave. fitness of population

Genetic Load = 9-6 =3